Saaketh Reddy Chintha Reddy

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Education

Georgia Institute of Technology

MS Mechanical Engineering; GPA: 4.00 — May 2025 – May 2027

BS Mechanical Engineering; GPA: 3.90 — Aug 2020 – May 2025

Relevant Coursework: Nonlinear Controls, Convex Optimization, Linear Controls, Dynamics of Mechanical Systems, Robotics, Optimal Control, Mechatronics, System Dynamics

Research

Bipedal Robots: Lidar Lab / Lunar Lab — Atlanta, Georgia

Advisors: Prof. Ye Zhao & Prof. Lu Gan — Aug 2023 – Present

- Developed an online continual learning method for uncertainty-aware velocity estimation in legged robots operating on unstructured terrain; preparing a first-author manuscript
- Designed a Temporal Convolutional Network (TCN) architecture to estimate body and contact states for legged robots from histories of proprioceptive measurements
- Developed a ROS 2 framework for G1 hardware testing, integrating InEKF, FAST-LIO2, RL controllers, and online training with low communication latency and CPU overhead
- Developed an open-source, differentiable C++/Python InEKF library with Pinocchio integration, supporting position- and velocity-based kinematic residuals and body and world frame corrections
- Engineered a gear-pulley-driven parallel gripper to precisely track and transmit servo torque for robot manipulation.

Environmental Fluid Mechanics Lab — Atlanta, Georgia

Advisors: Dr. Mohammad Mohaghar — Jan – Aug 2025

- Led an undergraduate team in designing and prototyping a magnetically actuated silicone soft robot for fluid analysis studies using rapid prototyping and microfabrication techniques
- Designed flux-concentrating elements to strengthen the magnetic field to polarize the robot, validating magnetic studies through FEMM simulations and eliminating the need for costly polarizers

SEAL Georgia Tech Research Institute — Smyrna, Georgia

Student Assistant – Undersea Sonar Signal Processing — Jan – May 2024

Time–frequency spectral optimization of underwater sonar acoustic returns for characterization using spectral descriptors — W. N. Newcomb, S. Reddy, A. Satish, and A. Medda

- Developed a MATLAB framework to optimize signal detection using the smoothed pseudo-Wigner-Ville distribution
- Benchmarked detection performance against alternative time–frequency representations to quantify improvements

Experience

Georgia Institute of Technology — Atlanta, Georgia

Graduate Teaching Assistant — Aug 2025 – Present

Led weekly lab and recitation sessions, office hours, and grading for undergraduate mechanical engineering courses.

- *Intro to CAD – ME 1670*: Sketching principles and SolidWorks
- *System Dynamics – ME 3017*: Laplace analysis, state-space modeling, and system stability

Auriga Space — Los Angeles, California

Mechanical Engineering Intern — Aug – Dec 2024

- Supported mechanical design, machining, electrical assembly, and testing at an electromagnetic launch startup, delivering custom tooling on short notice to meet overnight build deadlines
- Led rapid prototyping operations using FDM and SLA printers, designing and fabricating 20+ tooling and structural components; experimentally refined material selection, part geometry, print settings, and resin curing conditions to improve strength and accelerate in-house fabrication
- Developed coil fabrication tooling and refined winding, curing, and demolding processes to improve consistency and achieve dimensional tolerances on the order of thousandths of an inch
- Developed a novel test bench to characterize the minimum pulse width detectable by photodiodes; designed a light-collimating optical setup for tracking projectiles traveling over 2,000 m/s

Sandia National Labs (Q Clearance Applicant DOE) Livermore, California

R&D Intern – Telemetry & Handling Gear Engineering Summer 2023 & 2024

- Designed 3 unique fixtures for vibration testing of telemetry assemblies. Performed modal & structural analysis in Creo to ensure fixture integrity. Produced machine drawings with GD&T annotations
- Automated vibration & shock test data analysis using MATLAB, cutting work by 75% and saving 20+ hours weekly
- Partnered with Pelican™ to design and engineer a foam shipping container, safeguarding a multi-million-dollar telemetry assembly from impacts, vibrations, and ESD during transport
- Defined requirements and procedures for both thermal and structural evaluations for a classified gas detector, ensuring compliance with DoD specifications.

Flowers Invention Studio Georgia Tech

Prototyping Instructor Aug 2023 – Present

- Train 20+ students weekly on the use of 3D printers, waterjets, laser cutters, woodworking and metalworking tools while assisting peers in integrating design for manufacturing & assembly for their academic/personal projects

Projects

Convexifying Contacts in Factor Graph Smoothing for Legged Robots May 2026

- Implemented a convex relaxation of foot-contact switching, reformulating contact selection as a linear program
- Integrated an Invariant Extended Kalman Filter with a GTSAM factor graph smoother, improving base velocity estimation accuracy by 52% over fixed contact assumptions

Optimal Control and Trajectory Optimization Jan 2026

- Implemented multi-agent trajectory optimization in Julia using Hermite–Simpson direct collocation and IPOPT with collision avoidance, nonlinear dynamics, and boundary constraints
- Developed iLQR for nonlinear cart-pole swing-up and time-varying LQR tracking to compensate for model mismatch
- Built a log-domain interior-point solver for contact dynamics with backward Euler integration; solved quadruped single-leg balance using Gauss–Newton optimization with line search

Serial Robotic Arm May 2025

- Implemented full manipulator modeling and control pipeline using DH parameters, forward/inverse kinematics, and dynamics, validated through end-effector trajectory tracking
- Designed and tuned joint-space PID controllers with gravity compensation in Simulink and analyzed torque profiles
- Optimized robot workspace performance via manipulability ellipsoid analysis, applying gradient-based optimization to maximize manipulability along the task trajectory

Noise Cancelling Headphone - Capstone Jan – May 2025

- Designed and manufactured all mechanical components of a custom noise-canceling headphone system from concept through fabrication, coordinating across subteams to integrate PCB packaging, thermal considerations, and cable routing within compact enclosures
- Led acoustic-mechanical design of sealed earcups and internal chambers to maximize passive noise isolation and sound quality through material selection, venting strategy, and sealing geometry
- Iteratively developed a headband for comfort, flexibility, and durability through geometry tuning, material selection, and rapid prototyping

Gesture-controlled RC car Aug – Dec 2023

- Integrated an IMU with a Texas Instruments microcontroller and Arduino to capture hand gestures
- Programmed motor drivers and designed a custom chassis and drivetrain incorporating omnidirectional wheels
- Developed and tuned a PD controller to maintain constant velocity under varying loads and inclines